

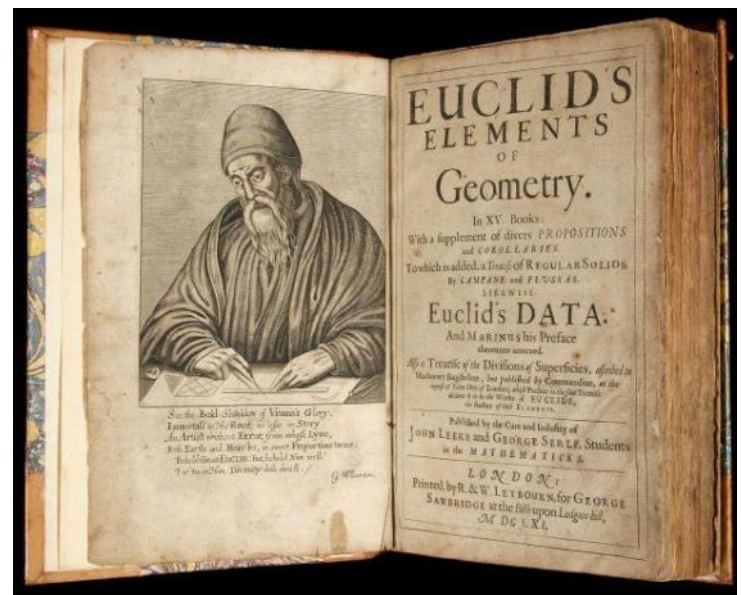
Geometry

Friday, May 17, 2024

11:44 AM

Euclid

- The Father of Geometry
- Lived 325 - 265 BC in Alexandria, Egypt
- Wrote "The Elements," his treatise of Geometry
- He defined many "axioms," i.e., statements considered true but are not proven

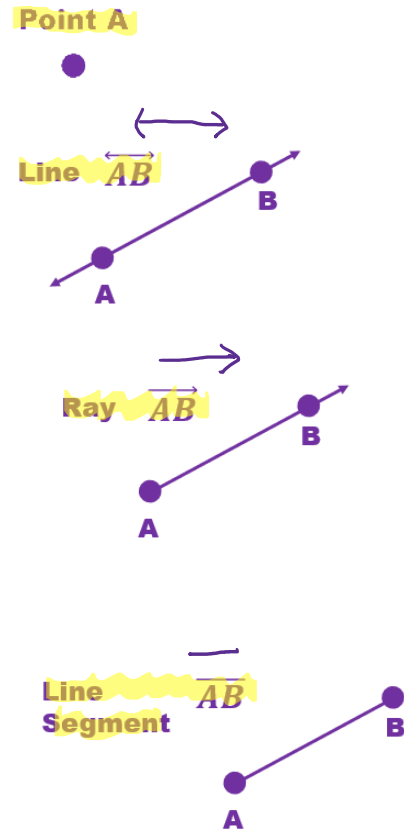


Axioms

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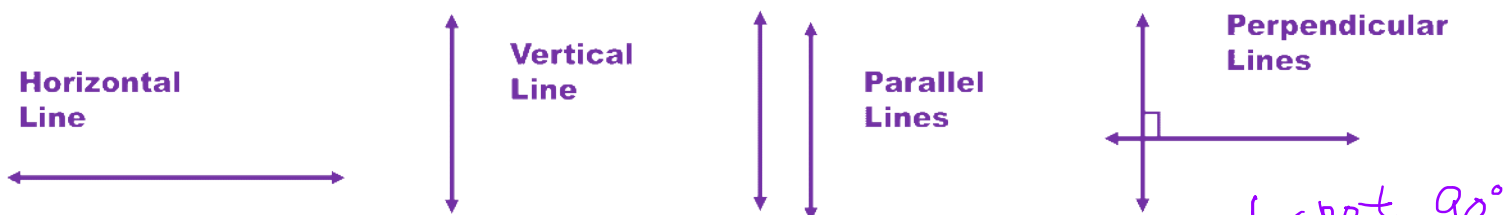
Some Axioms

- **Point** - that which has no part, no length, width, depth, but just a location in space. In Math, points are defined by a Capital letter (A, B, C, ...) or a Capital Letter with a numeric subscript (P_1 , P_2 , P_3 , etc.).
- **Line** - a straight path of points that continue infinitely in opposite directions. In Math, lines are defined by any two points through which the line passes with a double arrow line above the points ($\overleftrightarrow{AB}$). Sometimes, though unusual, we might see lines defined as a single small letter, such as p, q, r.
- **Ray** - a straight path of points with one terminal point and the other end that continues infinitely. A ray is also defined by two points, the terminal point and any other point, and a single arrow above the points ($\overrightarrow{AB}$).
- **Line Segment** - a straight path of points with two terminal points. A line segment is defined by its terminal points and a line above the points ($\overline{AB}$).



Lines

Saturday, May 18, 2024 7:49 AM



Answer the questions about the drawing to the right.

1. How many lines are shown in the drawing?

3 = p, q, and r

2. Point J lies at the intersection of perpendicular lines. Are lines r and p perpendicular? Explain your reasoning.

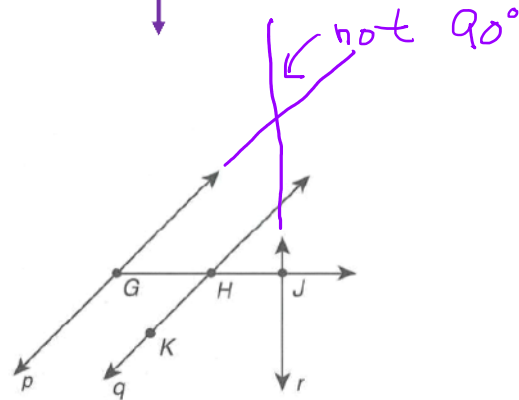
No. r & p are NOT perpendicular.
 Line & a RAY

3. Name 4 line segments found in the drawing.

$\overline{GH}$, $\overline{HJ}$, $\overline{GJ}$, $\overline{KH}$

4. There are many rays in the drawing. Name two.

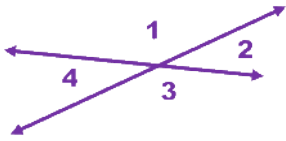
$\overrightarrow{GJ}$, $\overrightarrow{HJ}$



Angles

Friday, May 17, 2024 11:48 AM

Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.



Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Acute angle $\angle A < 90^\circ$ A diagram of an acute angle, which is less than 90 degrees. It is represented by two rays meeting at a vertex. A handwritten label "acute" with an arrow points to the angle.

Right angle $\angle A = 90^\circ$ A diagram of a right angle, which is exactly 90 degrees. It is represented by two rays meeting at a vertex, with a small square symbol at the vertex to indicate the right angle.

Obtuse angle $\angle A > 90^\circ$ A diagram of an obtuse angle, which is greater than 90 degrees. It is represented by two rays meeting at a vertex. A handwritten label "obtuse" with an arrow points to the angle.

Straight angle $\angle A = 180^\circ$ this is a straight line A diagram of a straight angle, which is 180 degrees. It is represented by a horizontal line with arrows at both ends, indicating it extends infinitely in both directions.

Complementary angles $\angle A + \angle B = 90^\circ$ A diagram showing two adjacent angles, A and B, that together form a right angle. A square symbol is placed at the vertex where the two angles meet, indicating that their sum is 90 degrees.

Supplementary angles $\angle A + \angle B = 180^\circ$ A diagram showing two adjacent angles, A and B, that together form a straight line. The rays of the two angles are collinear, forming a straight line.

These are just examples; there are an infinite number of angles

Angles, cont'd...

Monday, May 27, 2024 6:53 PM

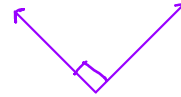
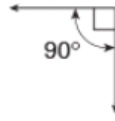
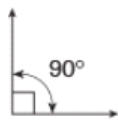
Name of Angle **Properties**

Examples

OR (for example)

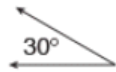
right angle

exactly 90°



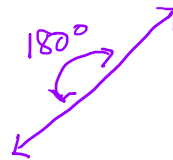
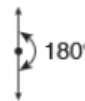
acute angle

less than 90°



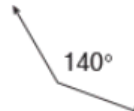
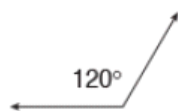
straight angle

exactly 180°



obtuse angle

between 90° and 180°



don't worry about this!!

Note: an angle of 225° would be called -135° obtuse.

Answer each question.

5. Match the illustrations with the descriptions.

D an acute angle

C a right angle

B an obtuse angle

A a straight angle

is this a Right angle?
we do NOT know!!
no square in the corner!!

a.

b.

c.

d.

6. Which of the three angles is smallest? Why?

(Hint: The size of an angle is based only on the angle, not the length of the sides.)

angle a because it is the only acute angle.

a.

b.

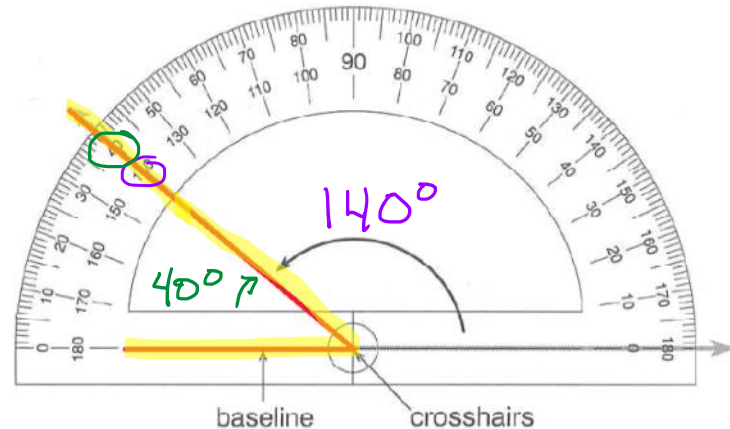
c.

USING PROTRACTORS

A **protractor** is a tool for measuring angles. A protractor looks like a half circle with a fanlike scale of numbers. One scale runs from left to right; the other from right to left.

To measure an angle, line up one ray of the angle with the baseline of the protractor. Put the center point of the protractor (sometimes marked with crosshairs) on the vertex of the angle. Read the scale at the point where the other ray crosses it.

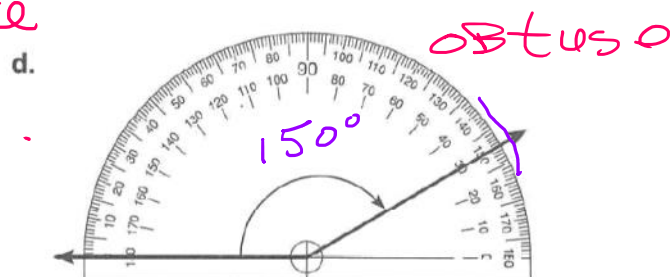
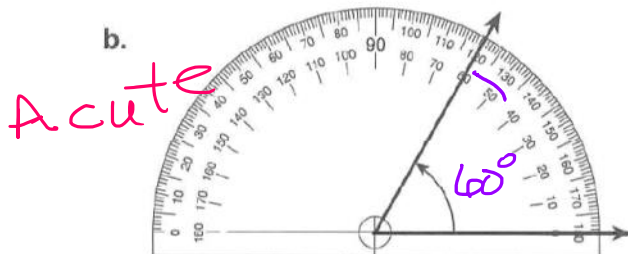
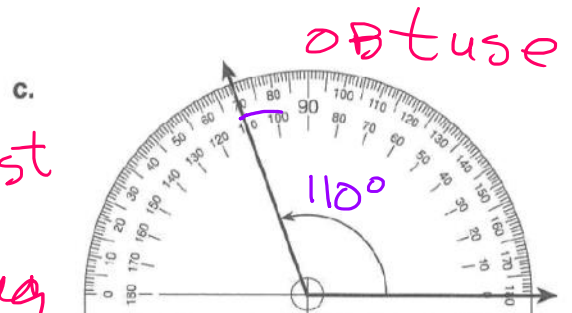
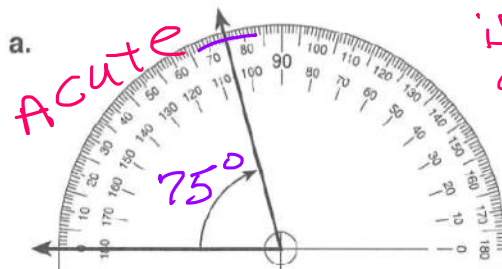
This illustration shows a protractor and an angle of 140° . The symbol $\angle$ means angle. Angles can be named by the vertex point, by a letter or number within the opening, or by three points.



Usually, $\angle A$ or $\angle B$ or ...

Answer each question.

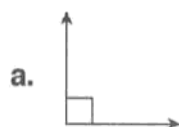
1. Tell the number of degrees in each of the following angles.



if you are not sure, just pick depending on acute or obtuse.

2. Match the angle with the correct number of degrees. Use your judgment or a protractor if you have one.

- b 180°
- a 90°
- c 30°
- d 135°

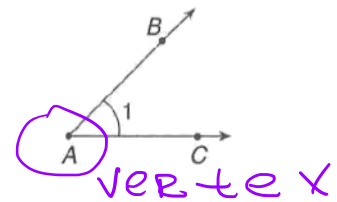


Naming Angels & Arithmetic

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The angle at the right could be named $\angle A$, $\angle 1$, $\angle BAC$, or $\angle CAB$. When three points are used to name an angle, the vertex is always the middle letter.

$\angle A$ OR $\angle BAC$ OR $\angle CAB$
 vertex

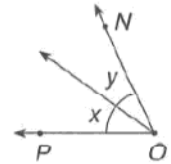


In this illustration, $\angle NOP$ is divided into two smaller angles, $\angle x$ and $\angle y$. If $\angle x = 35^\circ$ and $\angle y = 30^\circ$, we can add to find the measure of $\angle NOP$.

$$35^\circ + 30^\circ = 65^\circ$$

$$\angle NOP = 65^\circ$$

$$m\angle NOP = m\angle x + m\angle y$$



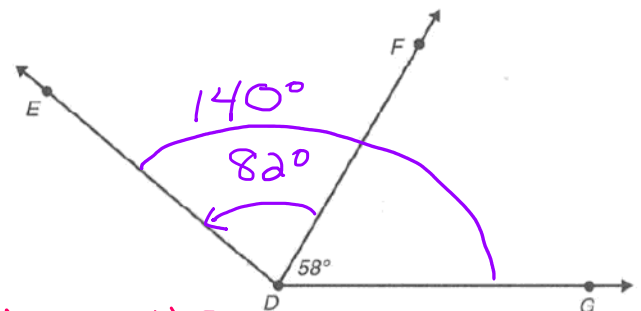
Subtraction can be used to find a missing angle.

(Note: The m before the angle symbol $\angle$ means "the measure of.")

$$m\angle EDG = 140^\circ. \text{ Find } m\angle EDF.$$

$$m\angle EDG - m\angle FDG = m\angle EDF$$

$$140^\circ - 58^\circ = 82^\circ$$

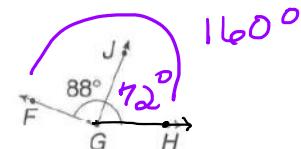


Notice: Everyone forgets to write "m" for "measure of".
 Solve the following problems without using a protractor.

3. $\angle FGH = 160^\circ$. Find the measure of $\angle JGH$.

$$\angle JGH = \angle FGH - \angle FGJ$$

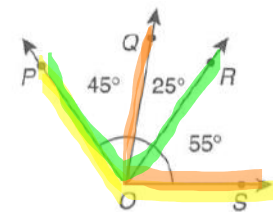
$$= 160^\circ - 88^\circ = 72^\circ$$



4. Tell the type of angle and number of degrees for each angle:

Angle	Type	Degrees
$\angle POS$	Obtuse	$45^\circ + 25^\circ + 55^\circ = 125^\circ$
$\angle QOS$	Acute	$25^\circ + 55^\circ = 80^\circ$
$\angle POR$	Acute	$45^\circ + 25^\circ = 70^\circ$

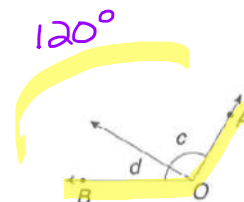
Hint: never trust a picture - determine degrees before identifying the type of angle.



5. $\angle AOB = 120^\circ$ and $\angle c$ is 3 times as great as $\angle d$. Use algebra to find the measurements of both $\angle c$ and $\angle d$. (Hint: Let a variable equal the measurement of $\angle d$.)

$$\angle d + \angle c = 120^\circ$$

$$\angle d + 3\angle d = 120^\circ$$



$$4 \angle D = 120^\circ$$

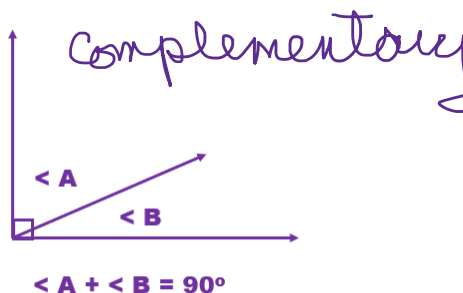
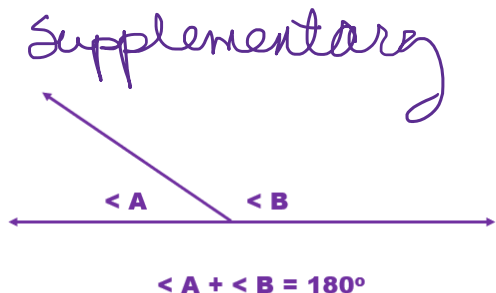
$$\angle D = 30^\circ \quad \& \quad \angle C = 90^\circ$$

Special Angles

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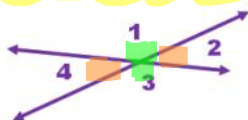
Recall:

- If $\angle A$ and $\angle B$ are Complementary Angles, then their measures add to 90°
- If $\angle A$ and $\angle B$ are Supplementary Angles, then their measures add to 180°



the box designates a right (90°) angle

Lines that intersect cross at exactly 1 point. The intersection of two lines creates 4 angles.

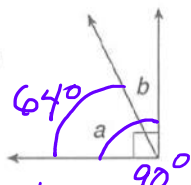


Vertical Angles are opposite each other $\angle 4$ & $\angle 2$ or $\angle 1$ & $\angle 3$

Solve as directed.

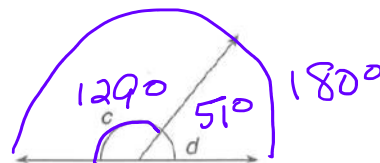
1. If $\angle b = 26^\circ$, what is the measure of $\angle a$?

$$\begin{aligned}\angle a &= 90^\circ - \angle b \\ &= 90^\circ - 26^\circ = 64^\circ\end{aligned}$$



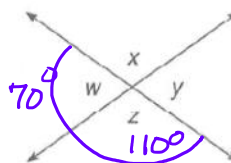
2. If $\angle c = 129^\circ$, what is the measure of $\angle d$?

51°



Use the drawing to answer questions 3–5.

3. Name two pairs of vertical angles. w & y ; x & z
4. If $\angle z$ measures 110° , what does $\angle w$ measure? 70°
5. $\angle z = 110^\circ$. What does $\angle x$ measure? How do you know?



Since $\angle z$ & $\angle x$ are vertical angles, $\angle x = 110^\circ$

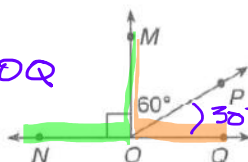
Use the drawing to answer questions 6 and 7.

6. Name a supplementary angle to $\angle MON$.

$\angle MOQ$

7. What is the measure of $\angle POQ$?

$\angle MOP$ & $\angle POQ$ are complementary angles, 90° total



Because

$$\angle MON + \angle MOQ = 180^\circ$$

also

$$\text{since } \angle MON = 60^\circ$$

~~angles~~
angles 90° total. $\rightarrow$ since $\angle MON = 90^\circ$
 $\angle POQ = 30^\circ$ Then $\angle MOQ = 90^\circ$

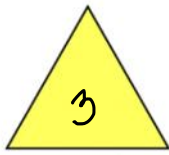
Polygons

Friday, May 17, 2024 11:49 AM

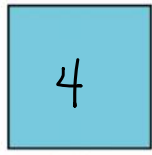
2-DIMENSIONAL

Definition of a Polygon: a plane figure with at least three straight sides, i.e., figures that are made completely of straight lines and the entire figure is closed.

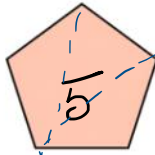
Polygons



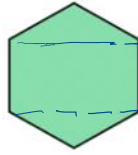
Triangle



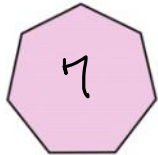
Quadrilateral



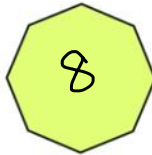
Pentagon



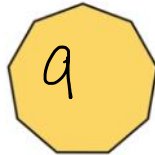
Hexagon



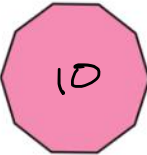
Heptagon



Octagon



Nonagon



Decagon

Note: it is redundant to say "straight line" because, by definition, all lines are straight.



what is the measure of each angle in:

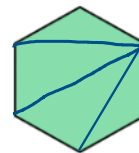
#3 $180^\circ \div 3 = 60^\circ$

#4 $90^\circ \times 4 = 360^\circ$

#5 $180^\circ \times 3 = 540^\circ$

#6 $(90^\circ \times 4) + (180^\circ \times 2) = 720^\circ$

Note it would be OKAY
hexagon as follows:



Hexagon

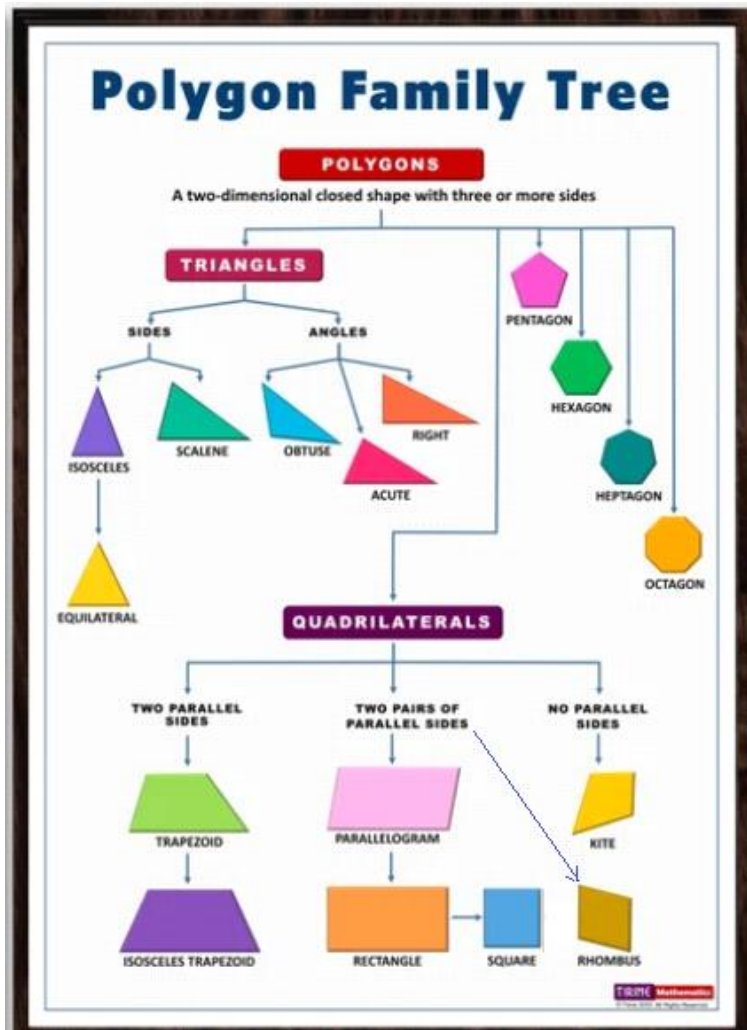
to slice the

$180^\circ \times 4 = 720^\circ$

The rest of the figures
are left to the students'
imagination!!

Polygon Family Tree

Tuesday, August 27, 2024 9:31 AM



note: a Rhombus is a parallelogram with a pair of opposite acute angles & a pair of opposite obtuse angles.

Quadrilaterals

Wednesday, May 29, 2024 7:58 PM

Definition of a Quadrilateral: a polygon with exactly 4 sides. The sum of the inside angles of a quadrilateral is 360° . The points where each pair of lines meet is called a vertex. Quadrilaterals all have exactly 4 vertices.

Definition of a Parallelogram: a parallelogram is a quadrilateral whose opposite sides are parallel and of equal length, and opposite angles are equal in measure.

Redundant Ditto

note: not possible to have opposite sides parallel and NOT be of equal length

Definition of a Rectangle: a rectangle is a parallelogram whose opposite sides are parallel and whose inside angles are all 90° each.

Definition of a Square: a square is a rectangle whose sides are all equal.

notation means Congruent.



Parallelogram



Rectangle



Rhombus



Square



Trapezium (UK)
Trapezoid (US)



Kite

These are ALL Quadrilaterals

Quadrilaterals, cont'd...

Wednesday, May 29, 2024 8:06 PM

4 sides, 4 vertices, Sum of Angles is 360°

Below are examples and properties of five common quadrilaterals.

Name	Properties	Examples
square	Opposite sides are parallel. All sides are equal. All angles are right angles (90°). Diagonals are equal and perpendicular.	
rectangle	Opposite sides are equal and parallel. All angles are right angles (90°). Diagonals are equal but not necessarily perpendicular.	
rhombus	Opposite sides are equal and parallel. All sides are equal. Diagonals are perpendicular but not necessarily equal.	
parallelogram	Opposite sides are equal and parallel.	
trapezoid	Only one pair of sides is parallel.	

$\perp 90^\circ$ angle

— congruent (equal)

chart is missing
KITE

Definition of a Kite: quadrilateral, 2 prs of adjacent equal sides, 2 prs of equal opp angles

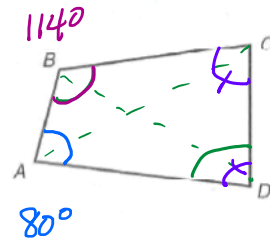


Quadrilaterals, Examples

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Use figure $ABCD$ to answer the questions below.

1. Which side is opposite AB ? CD
2. Name two sides that are adjacent to BC . AB & CD
3. Name the two diagonals. AC & BD
4. Which angle is opposite $\angle ABC$? $\angle ADC$



5. The measure of $\angle DAB$ is 80° . The measure of $\angle ABC$ is 114° . The remaining two angles are equal in measure. What is the measure of $\angle ADC$?

$$\angle ADC = 83^\circ$$

$$\begin{aligned} 114^\circ + 80^\circ + 2x &= 360^\circ \\ 2x &= 166^\circ \\ x &= 83^\circ \end{aligned}$$

check:

$$\begin{array}{r} 114 \\ 80 \\ \hline 166 \end{array} \quad \checkmark$$
$$\begin{array}{r} 166 \\ 166 \\ \hline 360 \end{array}$$

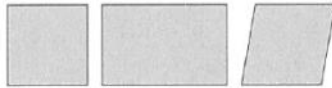
Note: $\parallel$ means parallel
 $\perp$ means perpendicular

Use the properties of quadrilaterals to answer each question.

6. Name three properties that rectangles and squares have in common.

① opp $\parallel$ sides ③ opp sides equal
 ② all angles 90° ④ diagonals equal

7. **True or False** A square, a rectangle, and a rhombus are three examples of parallelograms. Explain your answer.



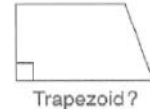
Yes, opp sides are $\parallel$ and equal.

8. A four sided figure has equal and parallel opposite sides. None of the four angles are right angles, but all four sides are equal in length. What is the name of the figure?



Rhombus

9. **True or False** A trapezoid can never have a right angle. Explain your answer. False. Def is 1 pr of $\parallel$ sides. that does not preclude a 3^{rd} side $\perp$.

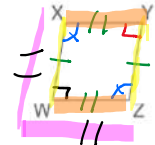


Trapezoid?

NOTE

- to prove False, need only 1 example
 - to prove True, must show for ALL cases

Imagine you are viewing the quadrilateral to the right from an angle that distorts the figure. You know that WX and YZ are equal, XY and WZ are equal, XW and WZ are equal, and $\angle XWZ$ is a right angle. Use the information to answer the questions below.



$\therefore$ all 90°

10. Is it possible for this figure to have only one right angle? No. Because $\parallel$ opp angles equal. Why or why not?

11. What is the name for this figure? square

12. **Explore** Tiles are often made in the shape of squares because it is easy to fit squares together to cover a surface. Would it be possible to use rhombus-shaped tiles to cover a surface? How about trapezoid tiles? Yes, except if the area is square or a rectangle, the corners won't be covered.

Triangles

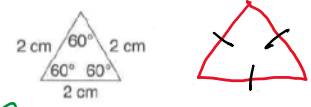
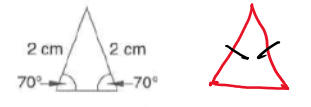
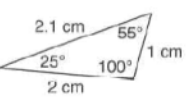
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Definition of a Triangle: a triangle is a polygon with three sides, and the sum of its inside angles is 180° .

NOTE:

① Spherical Geometry
- sum of angles $> 180^\circ$

② Hyperbolic Geometry
- sum of angles $< 180^\circ$

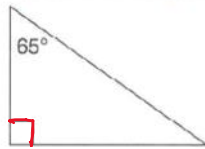
Name	Properties	Examples
equilateral used in TRIGONOMETRY to find sine & cosine of 30° & 60° angles	three equal sides three equal angles	
isosceles also used in TRIG to find sine & cosine of 45° angle	two equal sides two equal angles	
right	one right angle	
scalene	no equal sides no equal angles	

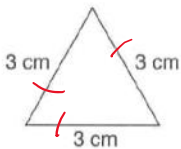
In BOTH cases, we restrict the lengths of the sides to make for easier calculations.

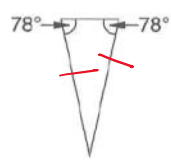
Triangles - Examples

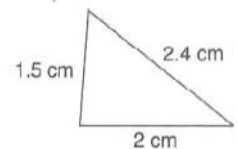
Thursday, May 30, 2024 10:56 AM

Identify each triangle.

- 

RIGHT
- 

EQUILATERAL
- 

ISOSCELES
- 

SCALENE

Use your knowledge of triangles to solve problems 5-7.

5. Determine which groups of three angles could form triangles.

30°, 30°, 90°

SUM of ANGLES
150°

✓ 50°, 80°, 50°

180°

✓ 45°, 60°, 75°

180°

120°, 30°, 45°

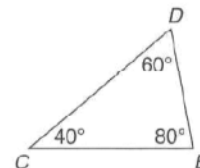
195°

✓ 50°, 60°, 70°

180°

6. What is the longest side in triangle CDE? Why?

CD Because it is opposite the largest angle *great question!!*



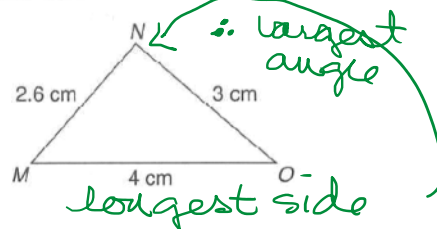
7. What is the largest angle in triangle MNO? How do you know?

$\angle N$

(OR, $\angle MNO$)

OR, $\angle ONM$)

Just a reminder!!



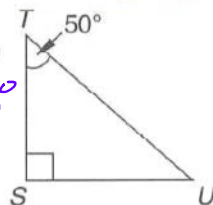
Use algebra to solve for the unknown angle(s).

8. What is the measure of $\angle U$?

$$90^\circ + 50^\circ + x = 180^\circ$$

$$x = 40^\circ$$

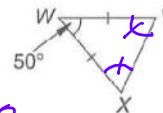
$$\angle U = 40^\circ$$



10. In the isosceles triangle VWX, $\angle W$ measures 50° . What are the measures of $\angle V$ and $\angle X$?

$$50^\circ + 2x = 180^\circ$$

$$2x = 130^\circ, x = 65^\circ$$



$$\therefore \angle V = \angle X = 65^\circ$$

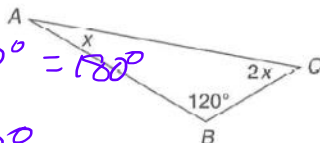
9. $\angle C$ is twice the measure of $\angle A$. If $\angle B$ measures 120° , what are the measures of $\angle A$ and $\angle C$?

$$x + 2x + 120^\circ = 180^\circ$$

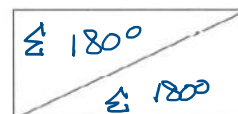
$$3x = 60^\circ$$

$$x = 20^\circ$$

$$\therefore \angle A = 20^\circ \text{ \& } \angle C = 40^\circ$$

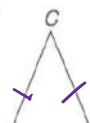


11. **Investigate** The rectangle below is divided into two triangles. Based on your knowledge of triangles, what is the sum of the angles in a rectangle? 360°



Σ This symbol means SUM (Greek capital Sigma)

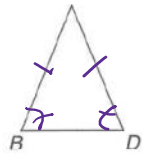
12. Triangle BCD is an isosceles triangle. If sides BC and CD are equal in length, then $\angle B$ must be equal to which other angle? $\angle D$



13. **Draw** Can there be a right angle in an isosceles triangle? Create a drawing that proves your answer.

yes!!

If sides BC and CD are equal in length, then $\angle B$ must be equal to which other angle? $\angle D$



triangle? Create a drawing that proves your answer.

